

Morphogenetic Events in Mixed Cultures of Rat Hepatocytes and Nonparenchymal Cells Maintained in Biological Matrices in the Presence of Hepatocyte Growth Factor and Epidermal Growth Factor

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Hepatocytes were grown in chemically defined hepatocyte growth medium (HGM) containing hepatocyte growth factor (HGF) and epidermal growth factor (EGF) on collagen-coated polystyrene beads in roller bottle cultures, forming clusters of beads, and proliferating hepatocytes and nonparenchymal cells, including fenestrated endothelium-forming vascular structures. Desmin-positive cells surrounded hepatocytes. Collagen types I and III were deposited in a diffuse manner whereas collagen type IV surrounded the clusters of the epithelial cells, forming a basement membrane. When the mixed cell clusters were implanted in Matrigel (Collaborative Research, Bedford, MA), hepatocytes grew in three dimensions, forming plates and ducts. Many single, long plates of hepatocytes were seen, suggesting progressive linear assembly guided by hepatocyte specific structural parameters. HGF, EGF, and transforming growth factor- α (TGF- α) enhance these phenomena. HGF plus EGF elicited maximal response. TGF- β 1 suppressed formation of the ducts and plates. Within three months in Matrigel, the cultures established monolayers composed of plates, ducts, and a well-delineated canalicular network. The mixed cultures expressed albumin, A1AT, AFP, transferrin, and CYP1B1. Following implantation of the cell clusters in Matrigel, there was decreased expression of c-met, urokinase, urokinase receptor, and TGF- β 1. Electron microscopy showed differentiated hepatocytes with nearly normal ultrastructure. The proliferating cell nuclear antigen (PCNA) labeling index was high (more than 80%) whereas the Bromo-deoxyuridine labeling index of ongoing DNA synthesis varied from 10% to 15%. These results show that the mixed cultures of proliferating hepatocytes and nonparenchymal cells can reproduce the hallmark structures of

hepatic histological architecture while maintaining differentiation and the capacity to proliferate. (HEPATOLOGY 1999;29:90-100.)

Hepatocytes are highly differentiated cells capable of multiple cycles of proliferation. Recent work with transgenic models indicates that the replicative capacity of non-neoplastic differentiated hepatocytes *in vivo* far exceeds limits established for fibroblasts in culture.¹⁻⁴ Serial transplantation of mature hepatocytes was able to rescue mice and repopulate livers for up to eight generations of mice. Normal hepatic architecture was restored when livers were repopulated. The parameters controlling morphogenetic events relevant to the formation of hepatic architecture, however, have not been defined.

Several laboratories have introduced culture techniques that allow hepatocytes in primary culture to grow and/or express complex patterns of differentiation.⁵⁻⁷ We also have recently established conditions that allow mature hepatocytes to enter into clonal expansion in cell culture.⁷ Hepatocytes cultured in chemically defined hepatocyte growth medium (HGM) enter into DNA synthesis in response to polypeptide mitogens, notably epidermal growth factor (EGF), transforming growth factor- α (TGF- α), and hepatocyte growth factor (HGF). These mitogens induce multiple rounds of DNA synthesis and expansion of the cell population. The proliferating cells, however, lose most markers of differentiation while they retain expression of hepatocyte associated transcription factors HNF1, HNF4, and HNF3. These undifferentiated cells become differentiated hepatocytes after addition of matrix derived from the Engelbreth-Holm-Swarm sarcoma (Matrigel [Collaborative Research, Bedford, MA]). The same cells predominantly form ducts when placed in collagen type I gels in the exclusive presence of HGF. In this study, we examined the behavior of these proliferating hepatocytes when forced to grow, not as monolayers but in three dimensions, suspended in a biological matrix. Of the many conditions tested, the best responses were observed from mixed cultures of hepatocytes and nonparenchymal cells growing as clusters of cells and collagen coated plastic beads in roller bottles, subsequently placed in Matrigel. The purpose of this article is to describe the conditions used and the results obtained. The presence of mixed-cell populations and the deposition of matrix in the roller bottle cultures imparts phenotypic stability in terms of differentiated gene expression. The morphogenetic patterns formed in these cultures when placed in Matrigel suggest that hepatocytes alone have innate tendencies to form structures

Abbreviations: HGM, hepatocyte growth medium; EGF, epidermal growth factor; TGF- α , transforming growth factor- α ; TGF- β 1, transforming growth factor- β 1; HGF, hepatocyte growth factor; PBS, phosphate-buffered saline; GAPDH, glyceraldehyde phosphate dehydrogenase; PCNA, proliferating cell nuclear antigen.

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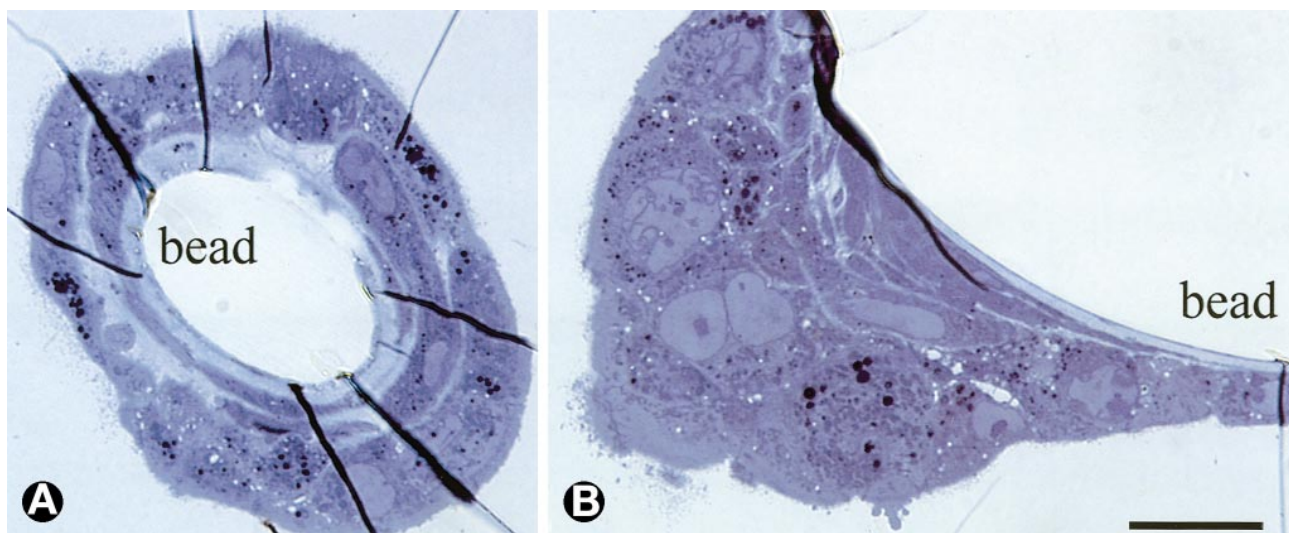


FIG. 1. Thin sections of cells on beads in roller bottle cultures at day 15 after isolation, stained with toluidine blue. (A) The bead is seen as a hollow space in the center of the cell cluster. Gray material around the bead represents dense type-I collagen deposition. The collagen surrounds and embeds connective-tissue-derived nonparenchymal cells. Cells with hepatocyte morphology surround the connective tissue core. (B) Similar as in A. The epithelial cells with hepatocyte morphology form an eccentric growth over a foundation of connective tissue cells. Note the formation of multiple microvilli over the hepatocytes present on the surface.

such as ducts and plates separated by sinusoidal-like spaces, as seen in the organization of the hepatic acinus.

MATERIALS AND METHODS

Animals

Male Fischer 344 rats from Charles River were used for the studies described. All animals were treated according to protocols approved by the animal care institutional review board.

Materials

EGF was obtained from Collaborative Biomedical (Waltham, MA). Collagenase for hepatocyte isolation was obtained from Boehringer Mannheim (Mannheim, Germany). Vitrogen (from Celtrix Labs., Palo Alto, CA) was used for the construction of the collagen gels. General reagents were obtained from Sigma (St. Louis, MO). EGF and Matrigel (Collaborative Research) were purchased from Collaborative Biomedical. HGF used for these studies was the $\Delta 5$ variant and was kindly donated by Snow Brand Co. (Toshigi, Japan). Polystyrene beads coated with type I collagen were purchased from SoloHill Inc. (Ann Arbor, MI). Antibody sources: Mouse anti-rat ICAM (CD54) Pharmingen (San Diego, CA) (1:500);

rabbit anti-rat collagen I, Chemicon (Temecula, CA) (1:100); rabbit anti-rat collagen III, Chemicon (1:100); Mouse anti-desmin, Dako (Carpenteria, CA) (1:100); Mouse anti-rat monocyte/macrophage (ED-1) Serotec (Raleigh, NC) (1:500); Rabbit anti-rat Collagen IV, generous gift from Dr A. Martinez-Hernandez (1:100).

Isolation and Culture of Hepatocytes

Rat Hepatocytes. Rat hepatocytes were isolated by an adaptation of Seglen's calcium two-step collagenase perfusion technique⁸ as previously described from our laboratory.⁹ Typically, a 3% contamination with nonparenchymal cells is seen in this isolate.

Nonparenchymal Cell Fraction. Nonparenchymal cell fraction was defined as the cell pellet isolated from the supernatant of the first low-gravity centrifugation used to prepare hepatocytes. This fraction primarily contains cells of Ito, bile duct cells, and endothelial cells. Small hepatocytes are also present in this fraction, typically comprising 5% of the cells.

Roller Bottle Cultures

Freshly isolated hepatocytes were added to roller bottles (850 cm² surface) obtained from Falcon (Franklin Lakes, NJ). Each bottle

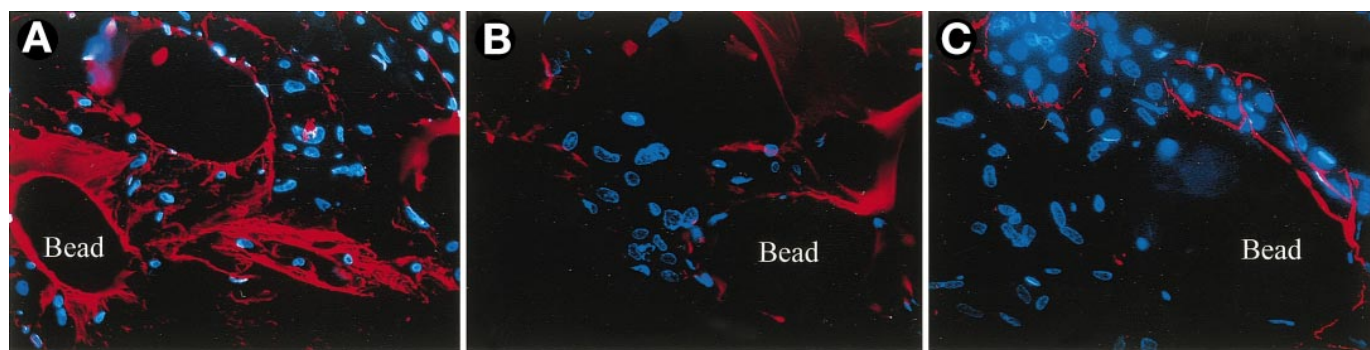
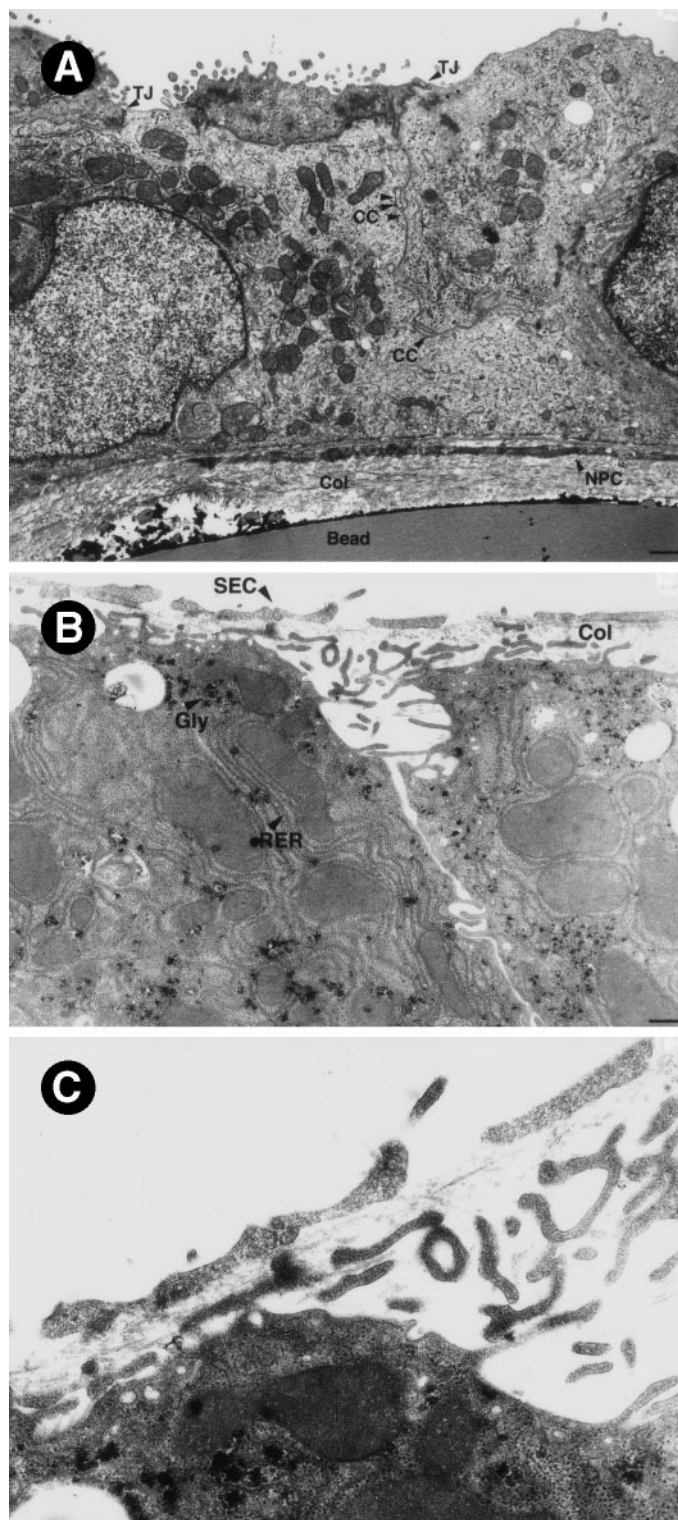


FIG. 2. Matrix deposition in Stage 1 roller bottle cultures. A, B, and C show depositions of collagen types I, III, and IV respectively. Collagen types I and III are deposited as broad bands surrounding the beads. Collagen type IV often formed basement membrane structures surrounding hepatocytes arranged in acinar or ductal configurations. Matrix is stained red whereas nuclei of the adjacent cells are stained blue. Visualization by immunofluorescence microscopy.

contained 18.7×10^6 beads and 210×10^6 freshly isolated hepatocytes in 250 mL of HGM medium supplemented with HGF (20 ng/mL) and EGF (10 ng/mL). The bottles were rotated at a rate of 2.5 rotations per minute and kept in an incubator maintained at 37°C, saturated humidity, and 5% CO₂. The viability of the cultures was assessed by periodic sampling. The samples were directly observed under a phase contrast microscope as well as stained with methyl tetrazolium to assess viability.



Cultures of Beads in Matrigel

The bead clusters containing cells were isolated from suspensions obtained from the roller bottle cultures. Enrichment for clusters was obtained by allowing for 2 minutes of unit gravity sedimentation. The bead and cell clusters were mixed with Matrigel (Collaborative Research). Bead clusters with cells were allowed to settle whereas beads without cells stayed mostly in suspension. The supernatant was aspirated leaving the clusters in the bottom of the tube. The process was repeated three times. Clusters suspended in medium were mixed with Matrigel at a volume ratio of 1:4 (medium plus beads: Matrigel). Approximately 50 to 100 bead clusters were randomly embedded in Matrigel.

Composition of the HGM

HGM was prepared as previously described.⁷ DMEM medium powder, HEPES, glutamine, and antibiotics were purchased from GIBCO/BRL (Grand Island, NY). ITS mixture (Insulin, Transferrin, Selenium) was purchased from Boehringer Mannheim. All other additives were cell-culture grade (Sigma). Unless otherwise indicated for specific experiments, the basal HGM consisted of DMEM supplemented with purified bovine albumin (2.0 g/L), glucose (2.0 g/L), galactose (2.0 g/L), ornithine (0.1 g/L), proline (0.030 g/L), nicotinamide (0.305 g/L), ZnCl₂ (0.544 mg/L), ZnSO₄·7H₂O (0.750 mg/L), CuSO₄·5H₂O (0.20 mg/L), MnSO₄ (0.025 mg/L), glutamine (5.0 mmol/L), and dexamethasone (10^{-7} mol/L). Penicillin and streptomycin were added to the basal HGM at 100 mg/L and 100 µg/L, respectively. The mixed basal HGM was sterilized by filtration through a 0.22-µm low-protein-binding filter system, stored at 4°C, and used within 4 weeks. ITS 1.0 g/L, (rh-insulin 5.0 mg/L, human transferrin 5.0 mg/L [30% diferric iron saturated], selenium 5.0 µg/L) was added after filtration immediately before use. The growth factors, as required, were added to HGM fresh at the specified concentrations every time the medium was changed.

Transmission Electron Microscopy

Samples for transmission electron microscopy were washed once in PBS with 1 mmol/L MgCl₂, 0.5 mmol/L CaCl₂, then fixed overnight at 4°C in 2.5% glutaraldehyde in PBS. Samples were washed three times with PBS then postfixed in 1% OsO₄, 1% KFe(CN)₆ in PBS for 1 hour at room temperature. Samples were washed three times in PBS, then dehydrated through graded series (30%-100%) of ethanol. Following three changes of 100% ethanol, samples were infiltrated with several changes of Polybed 812 resin (Polysciences, Warrington, PA) at room temperature, a change overnight at 4°C, then a final change, in the case of cells grown on monolayers, where Beem capsules, filled with resin, were inserted on top of areas of interest. Resin was hardened overnight at 37°C, then for 2 additional days at 65°C. While the resin was still warm, Beem capsules were pulled from the dish and analyzed to ensure that the

FIG. 3. Electron microscopy of cultures at Stage 1 (Roller bottle). (A) Low magnification view of hepatocytes growing on beads, before addition of Matrigel (Collaborative Research). Hepatocytes form a continuous multi-layer or monolayer culture around the beads and display circuitous, interdigitated cell-cell contacts within the abluminal membrane. Canalicular structures (CC) and tight junctions (TJ) are seen. A 1-micron thick layer of fibrillar collagen (Col) is evident between the hepatocytes' abluminal membranes and the polystyrene bead. A nonparenchymal cell (NPC) is also seen within the fibrillar collagen layer. Bar = 1 mmol/L. (B) Another view of the cytoplasmic features of hepatocytes at stage 1 (Magnification, 4,000×). Sinusoidal endothelial cells (SEC) are forming a layer of fenestrated endothelium. Fibrillar collagen (Col) and multiple microvilli are seen under the endothelial layer, with a morphology similar to that seen in the space of Disse. Glycogen (Gly) and lamellae of rough endoplasmic reticulum (RER) are seen in the cytoplasm of the adjacent hepatocytes. (C) Higher magnification of B (10,000×) showing the fenestrae of the endothelial layer. Collagen fibrils are seen in the interrupted cytoplasmic continuity of the endothelial cell at the site of the formation of the fenestra.

cells did not remain on the dish. In some cases monolayers were re-embedded to obtain cross sections. Thick sections (300 μ m), obtained using a Reichert (Vienna, Austria) ultramicrotome fitted with a diamond knife, were heated onto glass slides, stained with 1% Toluidine Blue, and rinsed with water. Ultrathin sections (60 nm) were collected on Formvar-coated (Fullam, Schenectady, NY) grids and stained with 2% uranyl acetate in 50% methanol for 10 minutes, then 1% lead citrate for 7 minutes. Sections were analyzed and photographed on a JEOL JEM 1210 transmission electron microscope at 80 kV.

Immunofluorescence Microscopy

Samples from roller-bottle cultures were fixed in 2% paraformaldehyde and 0.01% glutaraldehyde in PBS for 1 hour. Liver seeds were then stabilized by dipping them in 3% gelatin in PBS, then refixing them in the above fixative for an additional 15 minutes. Samples were incubated in 2.3 mol/L sucrose in PBS at 4°C overnight. Samples were mounted on screw stubs and snap-frozen in liquid nitrogen. Five hundred nanometer-thick frozen sections were cut on a FCS Ultracut Microtome (Reichert) fitted with a cryokit. Sections were attached to glass slides by adsorbed Cell-Tak (Collaborative Biomedical). Sections were washed in 0.5% BSA, 0.15% glycine in PBS (PBG buffer) three times to remove sucrose, then blocked with 5% goat serum in PBG buffer for 30 minutes. Sections were then stained with various antibodies (see later) in PBG buffer for 1 hour at room temperature, washed three times in PBG buffer then stained with Cy3-conjugated (goat antirabbit or antimouse) secondary antibodies (Jackson Immunolabs, Bar Harbor, ME) for 1 hour. Sections were washed three times with PBG buffer, then once in PBS. Nuclei were stained with 0.1 mg/mL Hoechst (bisBenzimide) for 30 seconds, washed twice with PBS, then mounted on slides with use of gelvatol (23 g polyvinyl alcohol 2000, 50 mL glycerol, 0.1% sodium azide to 100 mL PBS), and viewed on an Olympus Provis epifluorescence microscope (Olympus America, Melville, NY) also equipped for differential interference microscopy.

Analysis of Gene Expression by Northern Blots

Extraction of Total RNA and Messenger RNA From Cultures. Total RNA was extracted by use of RNazol B (BioTECX, Houston, TX). RNA extraction from roller-bottle cultures was performed by washing bead-cell clusters in phosphate buffered saline and further digestion of the clusters by adding an equal volume of Trypsin-Ethylenediaminetetraacetic acid (GIBCO-BRL) to the bead-cell suspension. The mixture was shaken at 37°C for 10 minutes. The bead-cell clusters were further washed in S+M buffer at 4°C three times. The bead-cell pellet was mixed with three volumes of RNazol and purified according to the manufacturer's guidelines.

RNA was extracted from Matrigel (Collaborative Research)-

embedded beads by vortexing using 2.0 mL of RNazol B (BioTECX) per 1 mL of beads in Matrigel and purified per the manufacturer's guidelines. RNA concentration and purity were determined by routine spectrophotometry. Size separation of 20 μ g RNA per lane was completed on denaturing 1% agarose gels and transferred to nylon membranes (Amersham, Piscataway, NJ) by the capillary method. After cross-linking under ultraviolet light, membranes were hybridized overnight with specific complementary DNAs (as indicated in Figure 12) that had been labeled with [α - 32 P]dCTP using an Amersham random primer kit. Membranes were subsequently washed under high stringency conditions and exposed to R film (photographic film) (Kodak, NY) for 1 to 3 days. Quantification of the RNA hybridization bands was performed by laser densitometry.

Sources of Complementary DNA Probes

EGF-R (rat) originated from Dr. Sheldon Earp, University of North Carolina (at Chapel Hill); acidic fibroblast growth factor receptor from American Type Culture Collection (catalog number 78222); acidic fibroblast growth factor receptor from American Type Culture Collection (catalog number 65796); urokinase plasminogen activator originated from Dr. Jay Degen, University of Cincinnati; cytochrome IIB1 from Dr. Steve Strom (University of Pittsburgh); complementary DNAs for albumin, α -fetoprotein were generated by Dr. Joe Locker (University of Pittsburgh).

RESULTS

Morphogenetic Events in Cultures of Different Stages

Stage 1: Cultures of Hepatocytes on Beads in Roller Bottles. Collagen-coated polystyrene beads, as described in Materials and Methods, were placed in roller bottles at a ratio of 18.7×10^6 beads to 210×10^6 freshly isolated hepatocytes. HGF and EGF were added as standard supplements in the HGM medium of the roller bottle cultures. Cells attached to the beads and, within 2 to 3 weeks, formed clusters of beads bound together with mesenchymal cells surrounded by layers of epithelial cells. The mesenchymal cells concentrate toward the center of the cluster and surround the individual beads (Fig. 1A and B). They are associated with heavy deposition of type I and type III collagen immediately against the surface of the bead (Fig. 2). The collagen bundles surround the mesenchymal cells. Collagen type IV was seen as a thin rim forming a basement membrane surrounding only acinar structures of epithelial cells. The epithelial cells grow outside of the mesenchymal cells and symmetrically surround the

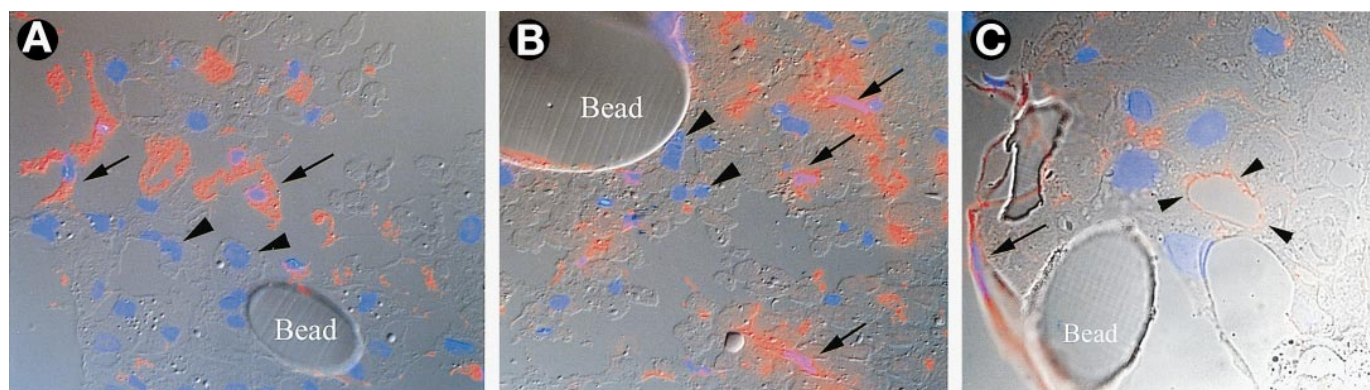


FIG. 4. Stains for macrophages, endothelial cells, and desmin-positive cells in Stage 1 roller bottle cultures. Visualization by differential interference microscopy. Positive immunohistochemistry is shown as red (complete arrows) whereas nuclei of cells are stained blue (truncated arrows). (A) Macrophages staining positive for ED-1 antigen. Note the "foamy" cytoplasm characteristic of macrophages in some of the cells. (B) Desmin-positive cells. (C) Structures of endothelial cells staining positive for ICAM1 antigen. One of the endothelial cells contains a nucleus at the field of the image (complete arrow).

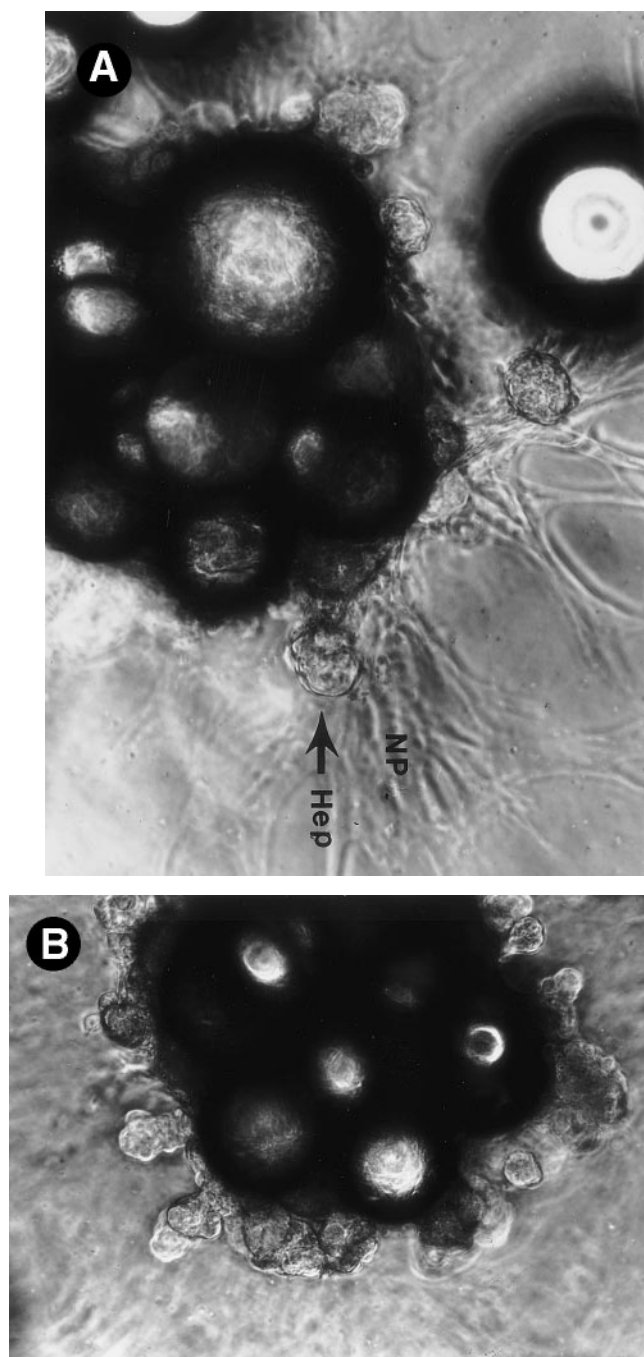


FIG. 5. Migration of cell populations from bead clusters after placement in Matrigel (Collaborative Research). Phase contrast microscopy. (A) Non-parenchymal cells (NP) migrate first and spread by attaching to the substratum. Occasional buddings of epithelial cells are seen at a higher focus plane (Hep). Some (arrow) appear to contain a duct. Culture at 1 week in Matrigel. Magnification, 200 $\times$. (B) Multiple buddings of epithelial cells migrate out of the bead clusters at different planes and in all directions. Culture at 20 days in Matrigel. Magnification, 200 $\times$.

beads or make eccentric projections. The epithelial cells have characteristics of small mature hepatocytes, as shown by electron microscopy. They contain multiple mitochondria and minimal rough endoplasmic reticulum (Fig. 3). Mature bile canaliculi containing microvilli as defined by junctional complexes were occasionally seen. Most often, they appeared as spaces surrounded by hepatocytes and containing micro-

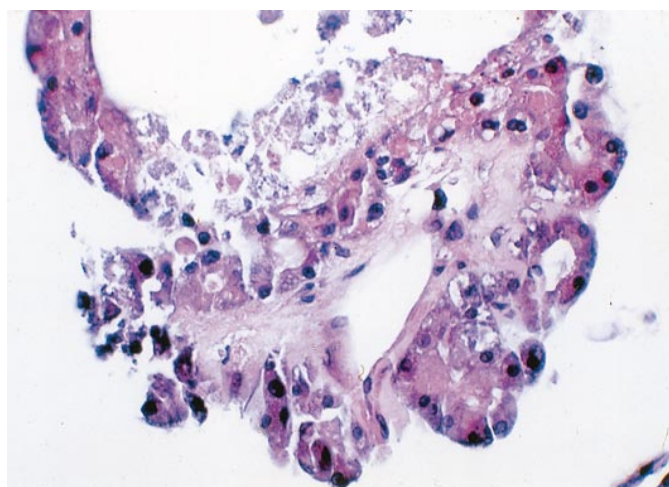
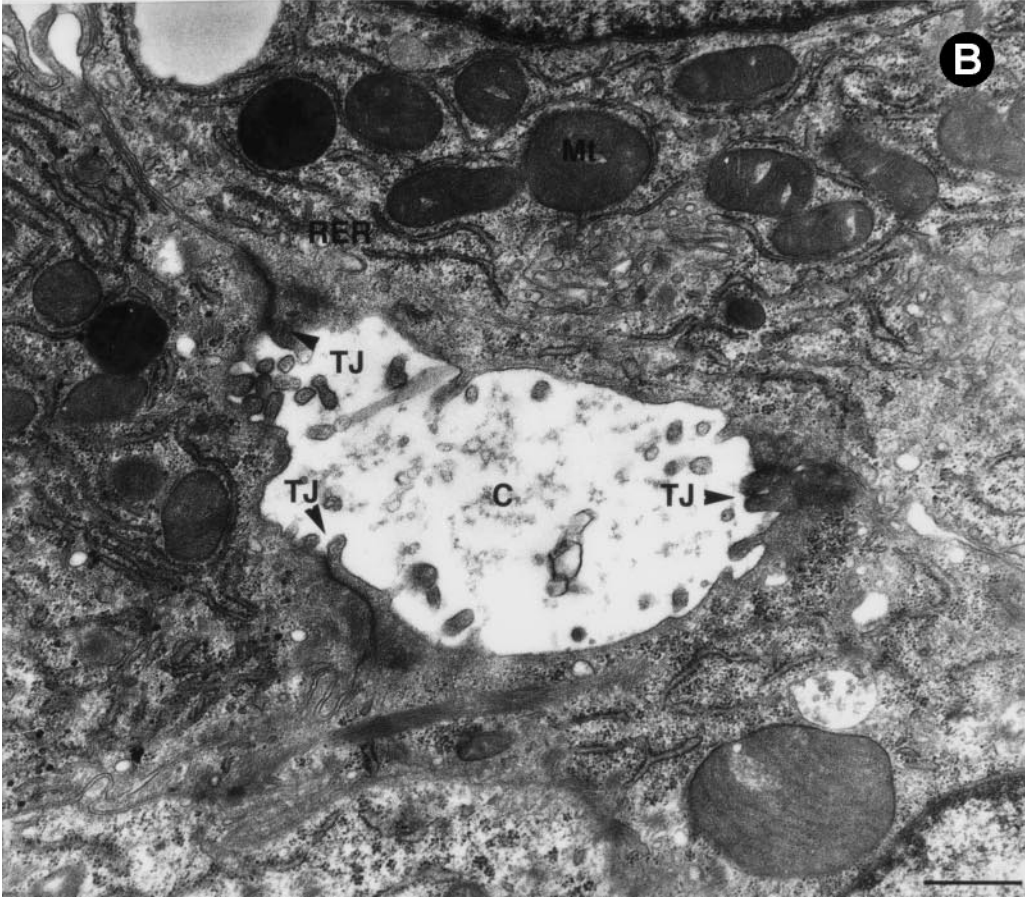
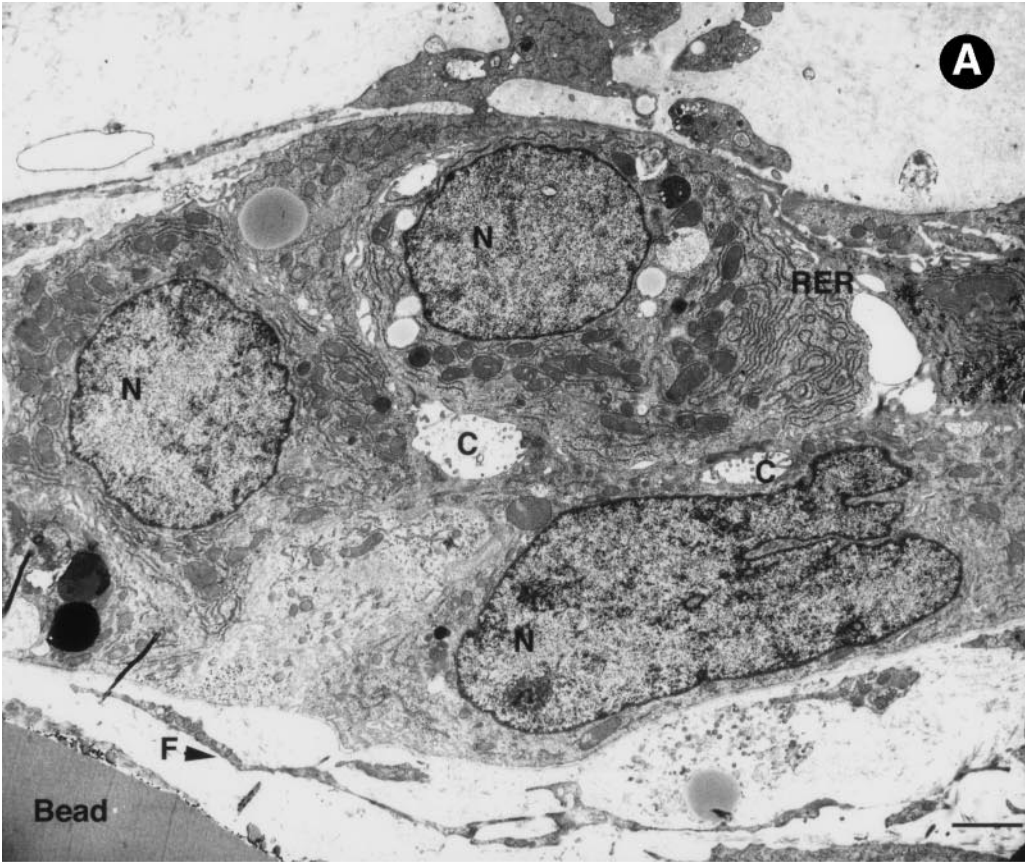


FIG. 6. Histology of the epithelial cell buddings in Matrigel (Collaborative Research) at Stage 2 cultures at day 20 in Matrigel. Epithelial cells with hepatocyte morphology (see Fig. 8) are surrounding the central bead core and are arranged in sheets and ducts. Connective tissue deposition is also present underlying the epithelial cell structures. Hematoxylin eosin stain. Magnification, 200 $\times$.

villi. The junctional complexes were not as clearly defined as after placement in Matrigel (Collaborative Research) (see later). Those cells that are on the surface of the clusters have visible microvilli, whereas those toward the interior do not. The epithelial cells form multiple cell layers from the mesenchymal cell layer of the cluster to the surface. The cytoplasmic details of the epithelial cells in the clusters are shown in Fig. 3B and C. Multiple lamellae of rough endoplasmic reticulum and glycogen deposition is seen. Notable is the occasional formation of fenestrated endothelium surrounding the hepatocytes. The PCNA labeling index of the epithelial cells exceeded 70% in all clusters. The BRdU labeling index of epithelial cells varied from 10% to 15% in different clusters. The number of nonparenchymal cells varied from one cluster to another. Figure 4 shows desmin-positive mesenchymal cells, presumably derived from stellate cells contaminating the original hepatocyte preparation,¹⁰ interspersed between the epithelial cells. Approximately 15% to 20% of the cells at this stage seem to belong to this category. ICAM1-positive endothelial cells are also seen in Fig. 4, occasionally forming ICAM1-positive luminal structures. Overall, less than 2% of the cells at this stage stained positive for this antibody. Macrophages, identified as ED-1-positive cells, are seen only in sporadic clusters, representing less than 0.1% of the total cell population.

Stage 2: Cultures in the First 3 Weeks After Implantation in Matrigel. Clusters of beads with the mixed cell populations were placed in Matrigel (Collaborative Research) as described in Materials and Methods. This resulted in a series of cell

FIG. 7. (A) Low power electron micrograph of an acinar structure formed from the bead cluster. Evident are the duct-like canalicular structures (C) in the center of the acinar structure. Cells contain extensive RER and numerous mitochondria. A thick, but less electron dense layer of extracellular matrix than that observed for the pre-Matrigel bead is seen between the hepatocytes and the bead, with several fibroblastic (F) type cells residing in the matrix. Bar = 2 μ m. (B) High power micrograph of the canalicular structure seen in A. Readily obvious are three extensive tight junctional areas (TJ), desmosomes, RER, Golgi elements, and Mt, mitochondria. Bar = 500 nm.



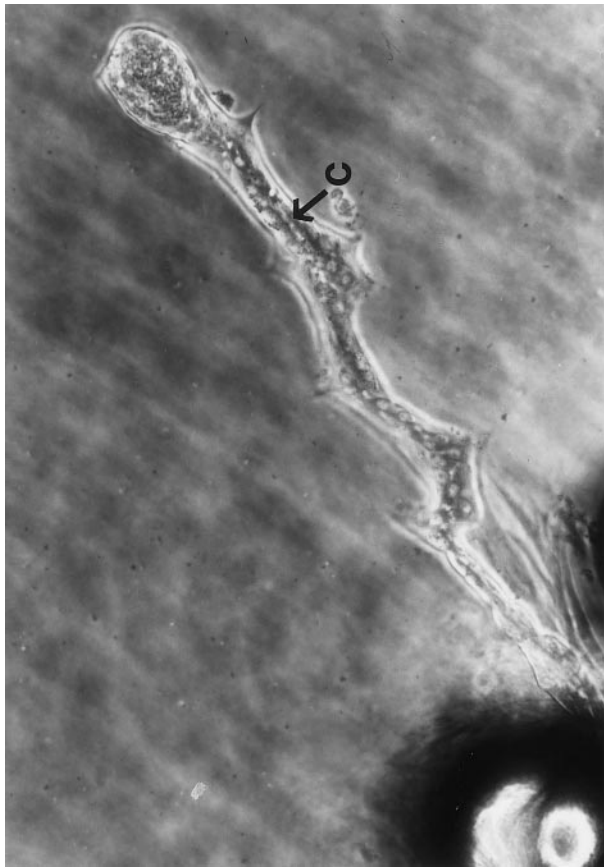


FIG. 8. Formation of plates by hepatocytes at Day 20 in Matrigel (Collaborative Research). Notice the prominent canalicular network (bright canals, arrows) along the middle of the plate.

migrations. Mesenchymal cells with stellate shape migrated out of the beads first at about day 4 to 5 and in many instances formed a mat surrounding the beads (Fig. 5A). Protrusions with rounded contours, appearing as buds, were seen extending randomly in all directions from the bead clusters at about day 7 to 10. Some of them (approximately 30%) appeared to contain ducts. The typical appearance of these cultures is shown in Fig. 5B. Sections of these bud structures stained with hematoxylin and eosin are shown in Fig. 6. The buds consisted primarily of hepatocytes arranged in acinar structures or in sheets. Electron microscopy (Fig. 7) showed enhanced cytoplasmic differentiation of hepatocytes compared with cells in the roller bottle. Hepatocytes in the buds contained abundant lamellae of rough endoplasmic reticulum, glycogen, and canaliculi with complete junctional complexes. The latter features are not seen in the hepatocytes before implantation in Matrigel. In most cultures, several long plates, 1 to 2 hepatocytes in width and 10 to 20 hepatocytes in length (Fig. 8), were seen. These structures averaged about 20 to 30 per plate, with plates of different length extending from most clusters. The plates typically developed into areas of the substratum that were free of other cell types. There were no visible nonparenchymal cells underlying or surrounding these plates. A typically demarcated and fully developed canalicular network was seen along the entire length of the plates. Many of these single plates contained ducts at the end. IL6 (10 ng/mL) added to the cultures augmented the number of duct structures and

caused formation of ducts along the plates or in the monolayer patches of hepatocytes. TGF- β 1 (at 0.5 ng/mL) inhibited formation of all structures that developed from epithelial cells (buds, plates, and ducts) though migration of the nonparenchymal cells was not inhibited. The full spectrum of changes was seen in the presence of HGF plus EGF. Cultures maintained in HGF or EGF alone showed fewer and more-limited changes per cluster compared with those with both growth factors. The extensive budding of the epithelial cells was associated with cell proliferation as judged by staining for PCNA. The numbers of labeled hepatocytes in the Matrigel ranged from 40% to 80% of epithelial cells per cluster, with considerable variation seen from one site to the next or among clusters. The BRdU labeling index, indicating active DNA synthesis, varied from 10% to 15% per cluster. Desmin-positive cells were seen interspersed and surrounding the hepatocytes. Type IV collagen was seen often as a thin rim surrounding acinar structures of hepatocytes. Slight staining was seen for type I and stronger staining for type III collagen (Fig. 9).

Stage 3: Long-Term Cultures in Matrigel. Long-term follow-up showed that HGF or EGF added separately was not sufficient to maintain prolonged viability of the epithelial cells. By 3 months, no epithelial cells were present in cultures maintained in HGF or EGF alone, or in control cultures without the addition of growth factors. In cultures maintained with combined HGF plus EGF, large monolayer patches of hepatocytes ranging from 2 to 10 mm in diameter were seen (Fig. 10). These structures appear at the rate of 2 to 4 patches per plate. These patches had a cytoarchitecture of striking similarity to sections of the liver acinus. Single or double hepatocyte plates were seen extending in a linear or convoluted manner. Complete canalicular networks developed throughout the entire length of each of the plates. The plates were separated by spaces that, though resembling the sinusoidal spaces seen in the liver lobules, did not contain any cells. Occasional ducts were also present in random locations along the plate structures. Electron microscopy (Fig. 11) showed typical hepatocyte morphology with most features typically present in hepatocytes, including glycogen, abundant rough endoplasmic reticulum, microbodies, and bile canaliculi with mature junctional complexes.

Gene Expression Changes in Cultures at Stages 1 and 2

The expression of several genes was examined in cultures at stages 1 and 2. Monolayers at stage 3 were not available in sufficient numbers for RNA preparation. Figure 12 compares expression of several genes in hepatocytes and nonparenchymal cells immediately after isolation from liver, cells from roller bottle cultures at day 13, cells from roller bottle cultures at day 25, and cell-bead clusters at 12 days after implantation in Matrigel (Collaborative Research) (day 25 after cell isolation). The first and last lanes show expression of the same genes respectively in hepatocytes and the nonparenchymal cell fraction, immediately after isolation from the rat liver. (Please note that several hepatocyte-associated genes are expressed in this fraction as a result of contamination by small hepatocytes, as discussed in Materials and Methods.) Though Matrigel-enhanced expression of α -fetoprotein, cultures in the roller bottles and in Matrigel maintained high expression of albumin. EGF-receptor expression decreased in culture, whereas HGF-receptor expression was maintained in roller bottles and in Matrigel, though

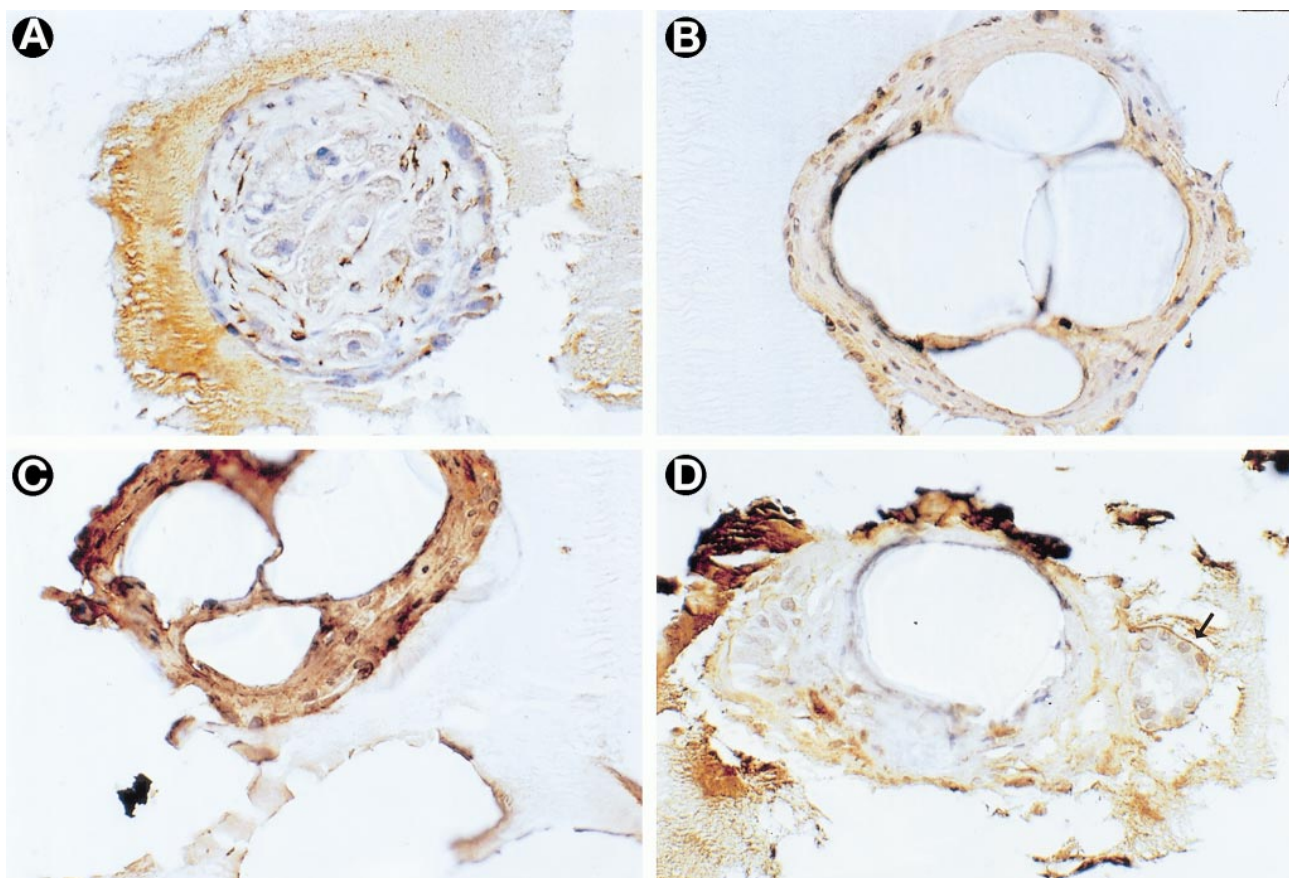


FIG. 9. Cellular and matrix immunohistochemistry in Stage 2 cultures in Matrigel (Collaborative Research). Staining by immunoperoxidase. A, B, C, and D show stains for desmin, Collagen types I, III, and IV respectively. Desmin-positive stellate cells are interspersed in close proximity to the hepatocytes. Collagen type III shows the strongest immunohistochemical response. Collagen type IV often formed basement membrane structures surrounding hepatocytes arranged in acinar or ductal configurations (arrow).

Matrigel caused a decrease in *c-met* expression. CYPB1 expression decreased gradually in the roller bottle cultures but was restored after addition of Matrigel. TGF- β 1 expression, derived from the nonparenchymal cells present in the mixed cultures, was pronounced in the roller bottle cultures at stage 1 but suppressed by Matrigel in stage 2 cultures. The same was true for urokinase plasminogen activator and its receptor urokinase plasminogen activator-R. Expression of transferrin and α -1 antitrypsin was also enhanced at stage 2. A separate study was conducted to evaluate induction of cytochrome P450 species in stage 1 cultures. Induction of cytochrome P450 species CYP1A, CYP3A, CYP2B1/2 was seen in response to 3' Methyl-cholanthrene, Dexamethasone, and Phenobarbital, respectively (data not shown).

DISCUSSION

Differentiation of hepatocytes in culture is dependent on the surrounding matrix, the chemical environment of the cell culture medium, and the presence of specific cytokines. Several previous studies have examined the differentiation and proliferation of hepatocytes in monolayers⁵⁻⁷ or spheroid cultures.¹¹ The system described here shows that mixed cultures of hepatocytes and nonparenchymal cells derived from liver maintain a remarkable degree of phenotypic stability in the mobile environment of roller bottle cultures. Typically, differentiation features such as induction of cytochrome P450 species and preservation of cellular morphol-

ogy are lost early on,⁷ unless hepatocytes are maintained in the presence of biomatrix gels such as Matrigel (Collaborative Research). Given the multiple examples with hepatocyte monolayer cultures, in which presence of matrix plays a defining role for hepatocyte differentiation, it is very likely that the phenotypic stability of hepatocyte functions in this system is dependent on the matrix synthesized by the nonparenchymal cells. This matrix directly surrounds hepatocytes arranged in acinar structures, as with collagen type IV (see Figs. 2 and 9) or provides a broader structural support as with collagen types I and III. Of interest is the relative abundance of the types of collagen seen in these studies. Collagens type I and III were most abundant whereas collagen type IV was present only in selected sites. This was seen in both Stage I and Stage 2 cultures. The proportions of the collagen species in these cultures parallels those seen in adult rat liver.¹²

The present study also shows that cells derived from mature hepatocytes can interact to develop many organotypic structures occurring in three dimensions during liver development, such as ducts, plates, and spaces between plates reminiscent of sinusoidal areas. The cells in these cultures are of two types, epithelial and mesenchymal. The epithelial cells are similar to those we have previously described in monolayer cultures maintained under the same conditions. We called these cells "hepatoblasts" because they are derived from hepatocytes by dedifferentiation and can redifferentiate to

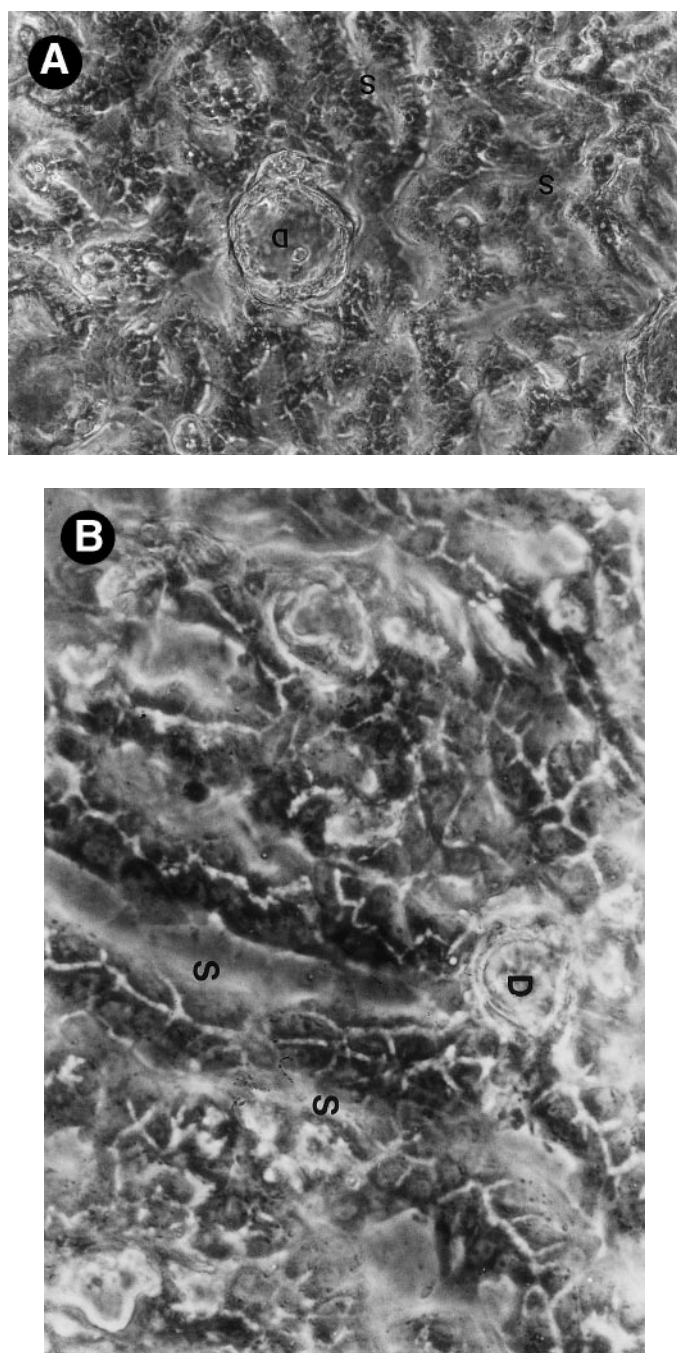


FIG. 10. Phase contrast microscopy of monolayers developing at 2 to 3 months in Matrigel (Collaborative Research) (Stage 3 cultures) in the presence of HGF and EGF. (A) Magnification, 100 $\times$. (B) Magnification, 200 $\times$. Notice the extensive canaliculus network (bright lines ramifying with short branches along the hepatocyte plates), the pseudo-sinusoidal spaces (S), and the duct-like structures (D).

form hepatocytes and cells organized in ductular structures. Given the fact that we use the same medium (HGM) and growth factors (HGF and EGF) as in the original study,⁷ we believe that these cells are the ones giving rise to the hepatocytes described in this study. The nonparenchymal cells observed in the current study however must derive from the contaminant 3% to 5% of cell population present from hepatocyte isolation.⁸ In standard monolayer cultures most of the nonparenchymal cells have been characterized as derived

from stellate cells.¹⁰ The stellate morphology of the nonparenchymal cells seen in our cultures and the positive immunohistochemistry for desmin also suggest the same derivation. Lipid droplets of different sizes were seen in these cells in Stage-2 cultures. On the other hand, endothelial cells in variable numbers are also present, as shown in Fig. 4. Given the microarchitecture of the cultures and the presence of Matrigel (Collaborative Research), the percentage of the Ito cells to endothelial cells cannot be precisely determined. They appear to be less than 15% of the overall population in Stage 1 and 2 cultures. Endothelial cells are much less than that (<1%), although sufficient for forming a fenestrated endothelium in conjunction with well-differentiated hepatocytes. Though occasional macrophage derived cells were also noted, their number did not exceed 2% of the total number of cells in these cultures. The mechanisms driving the formation of fenestrated endothelial cells have been studied by other investigators.¹³ The current model may allow definition of the fenestrated endothelial cell phenotype in relation to liver.

The presence of ducts and plates in Stage 2 shows that hepatocytes can form either of these structures while maintaining their phenotype. This is also seen *in vivo* in conditions such as fulminant hepatic failure in humans.¹⁴ Of interest were many single plates present in Stage 2, from 5 to 20 cells in length. Because these plates emerged from compact cell-bead clusters of Stage 1, it is likely that they evolved by gradual elongation caused by cell proliferation of either a basal cellular unit at the beginning of the plate or from all cells along the plate. The latter alternative seems more likely because PCNA-labeled cells were randomly placed along the plates. The finding of plate structures in culture suggests that this principle of forming plates is innate to hepatocytes and perhaps responsible for the linear assembly of hepatocytes within the liver acinus *in vivo*. In histological cross sections of normal liver, hepatocytes are arranged in plates surrounded by sinusoids. It is not clear whether this is also true in three dimensions or whether hepatocytes are in reality arranged in two-dimensional "walls," presenting as plates in cross sections. All plates, which were observed in this study, appeared attached to the substrate but were not in direct contact with nonparenchymal cells. The guiding principle leading to progressive elongation of the plates in a monodimensional linear pathway is not clear. Bile canaliculi were present along the entire length of the plates. These structures are fairly rigid, surrounded by a triple barrier of tight junctions, desmosomes, and gap junctions running along the entire length of the canaliculus. A sheath of contractile elements surrounds the above junctions and contributes to the maintenance of bile flow.¹⁵ We are proposing that the canaliculi establish rigid linear guides in the form of pipes, which link the progeny of cell division of single hepatocytes in such a way that a simple linear canaliculus is favored, free from angular distortion. This would tend to favor formation of linear plates traversed by a canaliculus network from end to end as seen in our cultures. It is quite possible that other structures on hepatocyte membranes besides canaliculi contribute to the formation of the linear plates. Such structures may be any of the multiple differentiated domains of hepatocyte membranes. It is important to underscore however that no mesenchymal cells appeared to surround or otherwise guide the plates. Hepatocytes by themselves provided the

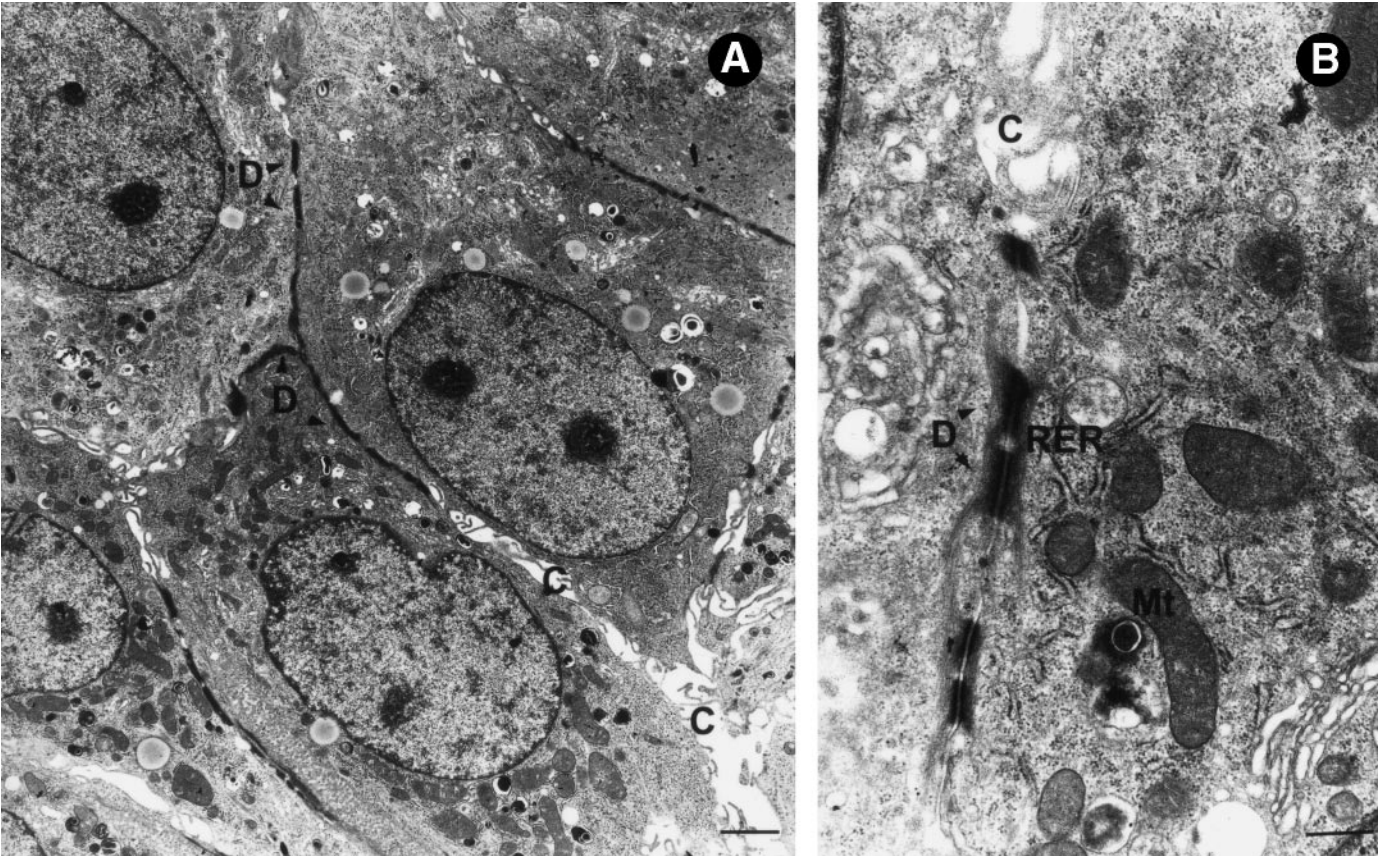


FIG. 11. (A) A low power (2,000X) electron micrograph of hepatocytes in Stage 3 cultures. Notice the longitudinal section of the extensive canalicular network (with microvilli and desmosomes) surrounding the individual hepatocytes. (B) Higher power view (10,000X) showing detailed cytoplasmic features. Rough endoplasmic reticulum, mitochondria, and Golgi network elements are seen in the individual hepatocytes.

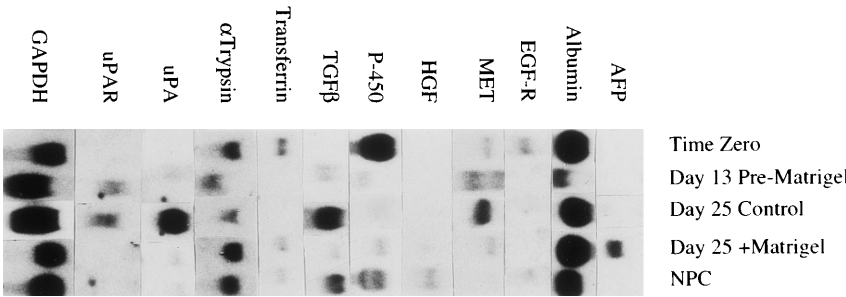
structural units that assembled these structures. This does not imply that nonparenchymal cells did not produce cytokines or matrix that affected this process; it suggests however that nonparenchymal cells are not directly involved in providing the structural components for the plate assembly. The model proposed from our results in Stage 3 cultures suggests that hepatocytes are responsible for the formation of the main microarchitecture of the acinus and that the stromal cells (Ito cells, sinusoidal endothelial cells, and Kupffer cells) fit within the architectural parameters established by the hepatocyte assembly.

The degree of differentiation seen in the roller bottle cultures was greater than previously observed in monolayer cultures under the same conditions. Roller bottle cultures as described in this study are easy to set up and maintain stability for long periods of time, exceeding 2 months (data

not shown). The stability of the cells on beads may be a result of the presence of multiple cell types, each providing matrix and cytokines for the adjacent cells. The stability of the cultures, the high degree of differentiation and the facility of harvesting make these cultures suitable for use in design of bio-artificial liver systems. Most of the systems used today rely on short-term cultures or established cell lines.¹⁶ The former lack longevity of function whereas the latter generally do not provide enough function to be of real use. Mass cultures as described for the Stage 1 in this presentation may have significant advantages in that they appear to overcome both of these deficiencies.

The combination of cell proliferation and differentiation seen in the cultures at Stage 2 with Matrigel (Collaborative Research) is unusual. Hepatocytes, as is the case with many other cell types, maintain differentiation but do not prolifer-

FIG. 12. Expression of several genes in hepatocytes immediately after isolation (Time zero), cells in roller bottle at day 13, cells in roller bottle at day 25, cells in Matrigel (Collaborative Research) cultures at day 25 (12 days after placement in Matrigel at Day 13), and nonparenchymal hepatic cell fraction (5% parenchymal hepatocyte contamination) immediately after isolation. Expression of GAPDH is used as a normalizing parameter. For comments, see Results.



ate in Matrigel.¹⁷ This has been variably ascribed to the presence of cytokines in Matrigel that inhibit cell proliferation, such as TGF- β 1.¹⁸ Indeed, commercial preparations of Matrigel depleted of TGF- β 1 have been made commercially available to overcome such antiproliferative effects.¹⁹ Our results indicate that TGF- β 1 is not responsible for inhibition of hepatocyte proliferation by Matrigel. In fact, more than 70% of hepatocytes were positive for PCNA in the presence of Matrigel, whereas addition of TGF- β 1 inhibited morphogenetic events and cell proliferation. If the low levels of TGF- β 1 present in Matrigel were responsible for the observed inhibition of hepatocyte proliferation seen in other systems, this does not appear to be the case here. Alternatively, it is also possible that hepatocytes in this system are for unknown reasons more resistant to TGF- β 1 and require higher concentrations than those present in Matrigel to cease cell proliferation. It should be noted that high expression of TGF- β 1 was seen in the roller bottle cultures presumably by the nonparenchymal cells. We do not know whether this generates active TGF- β 1 or the latent form or whether the concentration of TGF- β 1 in the high volume of the medium of the roller bottle cultures is sufficient to be effective.

The morphogenetic events of the epithelial cells were enhanced when both EGF and HGF were added to the cultures. Stage 3 cultures were not seen unless both EGF and HGF were present. We have previously shown that HGF and EGF have synergistic effects on the proliferation of monolayers of hepatocytes in HGM medium.⁷ The synergy between the two growth factors appears to also extend to events related to hepatocyte differentiation. A recent publication²⁰ showed that activation of EGF receptor by enhanced expression of TGF- α causes activation of the HGF receptor as well, suggesting a cross talk between the two different receptors. This cross talk may be responsible for the synergy between HGF and EGF observed in this study.

In summary, hepatocytes in long term roller bottle cultures (Stage 1) enter into a stable phenotype in terms of morphology and gene expression, probably because of the deposition of matrix by the associated nonparenchymal cells. The latter also form complex structures as evidenced by the fenestrated endothelium. Stage 2 cultures undergo complex morphogenetic transformations forming ducts and plates in the presence of HGF, EGF, and Matrigel (Collaborative Research). The growth in three dimensions forms ducts and sheets of mature hepatocytes surrounded by nonparenchymal cells. Proliferation and cell differentiation was very high in these cultures. Long plates of hepatocytes unattached to any other cell types were also seen, suggesting that hepatocytes alone have the innate tendency to form these structures. The capacity of hepatocytes in these Stage 2 cultures to undergo both sustained proliferation and maintain differentiation (in contrast to findings with cultures of pure hepatocyte monolayers) suggests that hepatic nonparenchymal cells are required for hepatocytes to be able to exercise both of these functions.

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